WIA1006/WID3006 MACHINE LEARNING

Semester 2, Session 2025/2026 | Universiti Malaya

COURSE INTRODUCTION

Course Synopsis

This course will covers supervised and unsupervised learning, including linear regression, classification, neural networks, and clustering, with hands-on coding using Python and popular libraries like Scikit-learn and TensorFlow. The course emphasizes both theoretical understanding and practical application, culminating in a final project where students apply ML algorithms to real-world datasets. It serves as a strong foundation for further study in machine learning and AI.

Credit:

- 3 Credit

Pre-requisite(s)

- None

Medium of Instruction

- English

Learning Outcome

At the end of the course, students are able to:

1. Understand and explain key concepts of machine learning algorithms.
2. Implement machine learning pipeline using Python and relevant libraries.
3. Evaluate machine learning performance using appriopriate metrics.

Course Topics

- Introduction to Machine Learning
- Data Preprocessing
- Regression Analysis
- Decision Tree/Random Forest
- Neural Network
- Bayes Theorem
- Clustering Algorithm
- Principle Component Analysis
- Support Vector Machine
- Gaussian Mixture Model
- Expectation Maximization Algorithm
- Reinforcement Learning
- Genetic Algorithm

Schedule

W	Date	Lecture (Monday)	Date	Tutorial (Tuesday, Wednesday, Thursday)	Remarks	Mode
1	09/03/2026	Introduction to Machine Learning	10/03/2026 11/03/2026 12/03/2026	Introduction to Machine Learning	**Nuzul Quran (07/03/2026)**	Lecture: Online F2F Tutorial: Online F2F
2	16/03/2026	Workshop - AWS CendekiAwan Malaysia	17/03/2026 18/03/2026 19/03/2026	**Workshop Reflection (4%)**	**Hari Raya Puasa (20/03/2026 - 21/03/2026)** ****Workshop Relection (4%)**	Lecture: Online F2F Tutorial: NF2F
3	23/03/2026	Data Preprocessing + Regression Analysis	24/03/2026 25/03/2026 26/03/2026	Data Preprocessing + Regression Analysis		Lecture: NF2F Tutorial: Online F2F
4	30/03/2026	Decision Tree/Random Forest	31/03/2026 01/04/2026 02/04/2026	Decision Tree/Random Forest	****Project Release (30/03/2026)**	Lecture: Physical F2F Tutorial: Physical F2F
5	06/04/2026	Neural Network	07/04/2026 08/04/2026 09/04/2026	Neural Network		Lecture: Physical F2F Tutorial: Physical F2F
6	13/04/2026	Case Study	14/04/2026 15/04/2026 16/04/2026	**Quiz (20%)**	****Quiz (20%)**	Lecture: NF2F Tutorial: Physical F2F
7	20/04/2026	Bayes Theorem	21/04/2026 22/04/2026 23/04/2026	Bayes Theorem		Lecture: Physical F2F Tutorial: Physical F2F
		SEM BREAK		**27/04/2026 - 03/05/2026**	**Labour Day, 01/05/2026**	
8	04/05/2026	Clustering Algorithm + Principle Component Analysis	05/05/2026 06/05/2026 07/05/2026	Clustering Algorithm + Principle Component Analysis		Lecture: Physical F2F Tutorial: Physical F2F
9	11/05/2026	Support Vector Machine + Gaussian Mixture Model	12/05/2026 13/05/2026 14/05/2026	Support Vector Machine + Gaussian Mixture Model		Lecture: NF2F Tutorial: Physical F2F
10	18/05/2026	ELITE Lecture	19/05/2026 20/05/2026 21/05/2026	**ELITE Lecture Reflection (4%)**	****Will appoint an ELITE for this course. Involvement in 3 weeks.** ****ELITE Lecture Relection (4%)**	Lecture: Physical F2F Tutorial: NF2F
11	25/05/2026	ELITE Lecture	26/05/2026 27/05/2026 28/05/2026	**ELITE Lecture Reflection (4%)**	**Hari Raya Haji (27/05/2026)** **Wesak Day (31/05/2026)** ****ELITE Lecture Relection (4%)**	Lecture: Physical F2F Tutorial: NF2F
12	01/06/2026	Expectation-Maximization Algorithm + Reinforcement Learning + Genetic Algorithm	02/06/2026 03/06/2026 04/06/2026	Expectation-Maximization Algorithm + Reinforcement Learning + Genetic Algorithm	**His Majesty the King's Birthday (01/06/2026)**	Lecture: NF2F Tutorial: Physical F2F
13	08/06/2026	**Project Presentation (10% + 4%)**	09/06/2026 10/06/2026 11/06/2026	**Project Presentation (10% + 4%)**	**UM IFest (08/06/2026 - 11/06/2026)** ****Project Submission (14%)**	Lecture: NF2F Tutorial: NF2F
14	15/06/2026	ELITE Lecture	16/06/2026 17/06/2026 18/06/2026	**ELITE Lecture Reflection (4%)**	**Awal Muharram (16/06/2026)** ****ELITE Lecture Relection (4%)**	Lecture: Physical F2F Tutorial: NF2F
15		Study week		**Final Exam (50%)**	****Final Exam (50%)**	Physical F2F

Assessment (100%)

- Week 2 – Workshop Reflection (4%) – NF2F
- Week 6 – Quiz (20%) – F2F
- Week 10 – ELITE Lecture Reflection (4%) – NF2F
- Week 11 – ELITE Lecture Reflection (4%) – NF2F
- Week 13 – Project Presentation (10%+4%) – NF2F
- Week 14 – ELITE Lecture Reflection (4%) – NF2F
- Exam Week – Final Exam (50%) – F2F

MS TEAMS

- General | WIA1006/WID3006 Machine Learning S2,25/26 | Microsoft Teams
- Team Code: axq45lt

GROUP PROJECT

- GROUP FORMATION
- Max 5-6 members per group. **Only can form within your own OCC.**
- Project Question will be released in Week 4.
- Project Presentation (NF2F), Submission Deadline: Week 13.

Subject to changes, reminders will be given from time to time.

Reference Books

- T V Geetha, S Sendhilkumar, *Machine Learning - Concepts, Techniques and Applications,* Routledge Taylor & Francis Group, 2023.
- Kevin P. Murphy, *Machine Learning: A Probabilistic Perspective*, MIT Press, 2021
- *Hands-on Machine Learning with Scikit-Learn, Hard, and TensorFlow: Concepts, Tools, and Techniques to Build Intelligent Systems 3rd Edition*, O'Reily Media, Inc, 2022
- Andrew Ng, *Machine Learning Yearning,* DeepLearning.AI, 2018
- Ian Goodfellow, Yoshua Bengio and Aaron Courville. *Deep Learning*. MIT Press 2016
- Yuxi (Hayden) Liu, Python Machine Learning by Example. Packt Publishing. 2019
- Andriy Burkov, The Hundred-Page Machine Learning Book. Andriy Burkov. 2019
- Giuseppe Bonaccorso, Mastering Machine Learning Algorithms. Packt Publishing. 2019

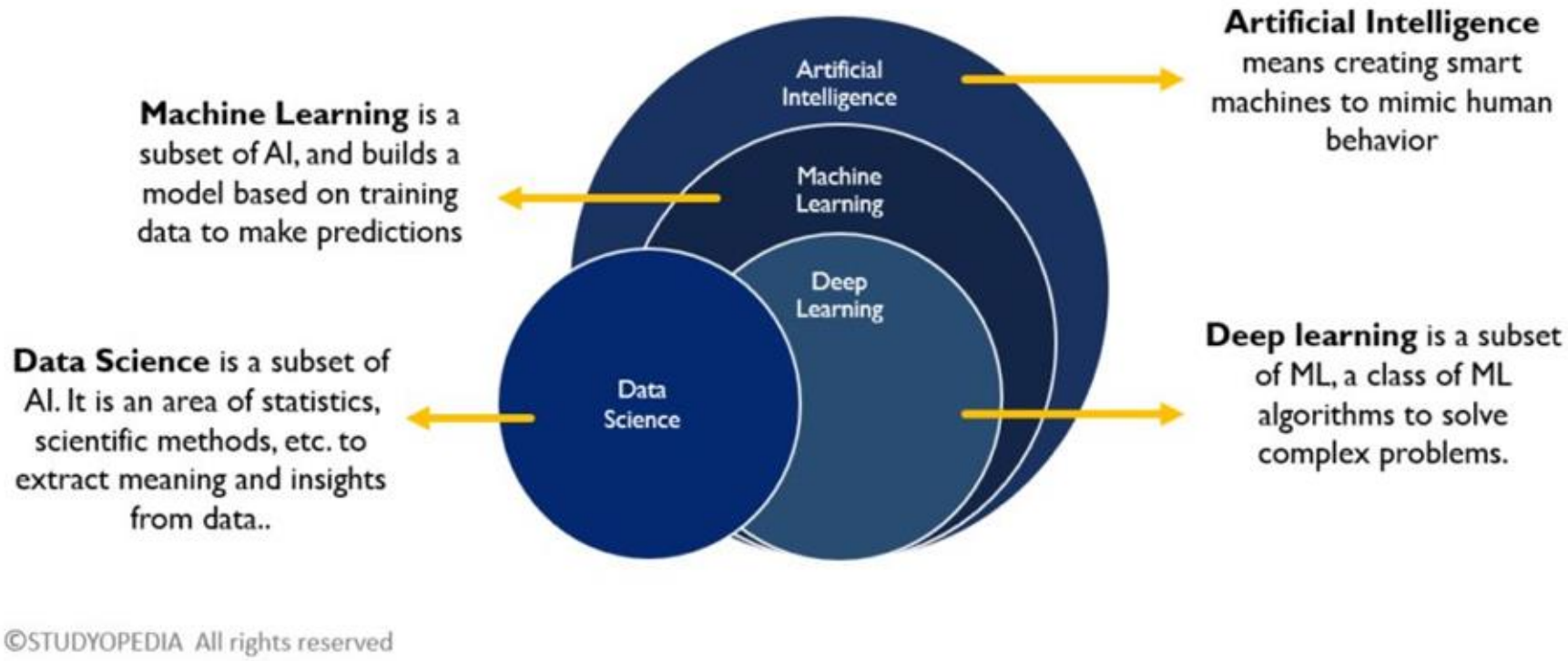
Q & A

INTRODUCTION TO MACHINE LEARNING

What is Learning?

- Herbert Simon: "*Learning is any process by which a system improves performance from experience.*"
- What is the task?
  - Classification
  - Problem solving / Planning / Control

Overview of AI



Overview of Machine Learning

- Supervised
  - Regression
  - Classification
- Unsupervised
  - Clustering
- Reinforcement Learning
- Self Supervised Learning

Classification

- Assign object/event to one of a given finite set of categories.
  - Medical diagnosis
  - Credit card applications or transactions
  - Fraud detection in e-commerce
  - Worm detection in network packets
  - Spam filtering in email
  - Recommended articles in a newspaper
  - Recommended books, movies, music, or jokes
  - Financial investments
  - DNA sequences
  - Spoken words
  - Handwritten letters
  - Astronomical images

Problem Solving / Planning / Control

- Performing actions in an environment in order to achieve a goal.
  - Solving calculus problems
  - Playing checkers, chess, or backgammon
  - Balancing a pole
  - Driving a car or a jeep
  - Flying a plane, helicopter, or rocket
  - Controlling an elevator
  - Controlling a character in a video game
  - Controlling a mobile robot

Measuring Performance

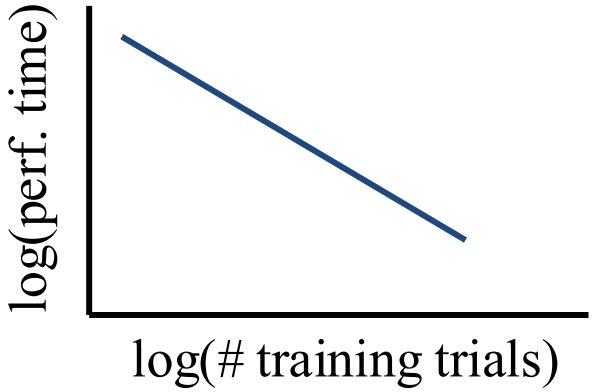
- Classification Accuracy
- Solution correctness
- Solution quality (length, efficiency)
- Speed of performance

Why Study Machine Learning? Engineering Better Computing Systems

- Develop systems that are too difficult/expensive to construct manually because they require specific detailed skills or knowledge tuned to a specific task (***knowledge engineering bottleneck***).
- Develop systems that can automatically adapt and customize themselves to individual users.
  - Personalized news or mail filter
  - Personalized tutoring
- Discover new knowledge from large databases (***data mining***).
  - Market basket analysis (e.g. diapers and beer)
  - Medical text mining (e.g. migraines to calcium channel blockers to magnesium)

Why Study Machine Learning? Cognitive Science

- Computational studies of learning may help us understand learning in humans and other biological organisms.
  - Hebbian neural learning
    - "Neurons that fire together, wire together."
  - Human's relative difficulty of learning disjunctive concepts vs. conjunctive ones.
  - Power law of practice



Why Study Machine Learning? The Time is Ripe

- Many basic effective and efficient algorithms available.
- Large amounts of on-line data available.
- Large amounts of computational resources available.

Related Disciplines

- Artificial Intelligence
- Data Mining
- Probability and Statistics
- Information theory
- Numerical optimization
- Computational complexity theory
- Control theory (adaptive)
- Psychology (developmental, cognitive)
- Neurobiology
- Linguistics
- Philosophy

Defining the Learning Task

Improve on task, T, with respect to performance metric, P, based on experience, E.

- T: Playing checkers
- P: Percentage of games won against an arbitrary opponent
- E: Playing practice games against itself

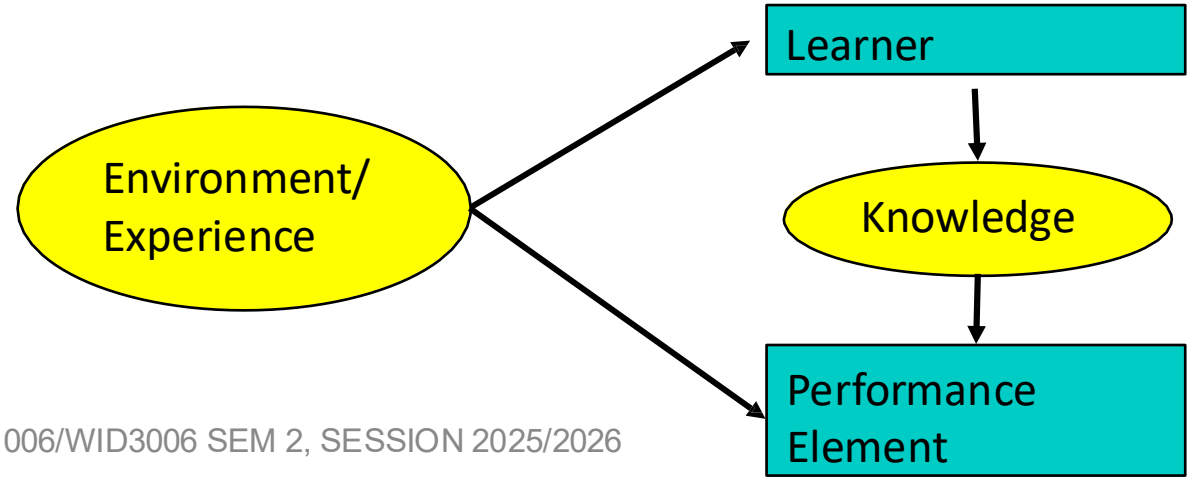
- T: Recognizing hand-written words
- P: Percentage of words correctly classified
- E: Database of human-labeled images of handwritten words

- T: Driving on four-lane highways using vision sensors
- P: Average distance traveled before a human-judged error
- E: A sequence of images and steering commands recorded while observing a human driver.

- T: Categorize email messages as spam or legitimate.
- P: Percentage of email messages correctly classified.
- E: Database of emails, some with human-given labels

Designing a Learning System

- Choose the training experience
- Choose exactly what is too be learned, i.e. the ***target function***.
- Choose how to represent the target function.
- Choose a learning algorithm to infer the target function from the experience.



Sample Learning Problem

- Learn to play checkers from self-play
- We will develop an approach analogous to that used in the first machine learning system developed by Arthur Samuels at IBM in 1959.

Training Experience

- Direct experience: Given sample input and output pairs for a useful target function.
  - Checker boards labeled with the correct move, e.g. extracted from record of expert play
- Indirect experience: Given feedback which is not direct I/O pairs for a useful target function.
  - Potentially arbitrary sequences of game moves and their final game results.
- Credit/Blame Assignment Problem: How to assign credit blame to individual moves given only indirect feedback?

Source of Training Data

- Provided random examples outside of the learner's control.
  - Negative examples available or only positive?
- Good training examples selected by a "benevolent teacher."
  - "Near miss" examples
- Learner can query an oracle about class of an unlabeled example in the environment.
- Learner can construct an arbitrary example and query an oracle for its label.
- Learner can design and run experiments directly in the environment without any human guidance.

Training vs. Test Distribution

- Assume that the training and test examples are independently drawn from the same overall distribution of data.
  - » IID: Independently and identically distributed
- If examples are not independent, requires
  - » **collective classification**.
- If test distribution is different, requires
  - » **transfer learning.**

Choosing a Target Function

- What function is to be learned and how will it be used by the performance system?
- For checkers, assume we are given a function for generating the legal moves for a given board position and want to decide the best move.
  - Could learn a function: ChooseMove(board, legal-moves) → best-move
  - Or could learn an ***evaluation function***, *V*(board) → **R**, that gives each board position a score for how favorable it is. *V* can be used to pick a move by applying each legal move, scoring the resulting board position, and choosing the move that results in the highest scoring board position.

Ideal Definition of *V*(*b*)

- If b is a final winning board, then V(b) = 100
- If b is a final losing board, then V(b) = –100
- If b is a final draw board, then V(b) = 0
- Otherwise, then V(b) = V(b´), where b´ is the highest scoring final board position that is achieved starting from b and playing optimally until the end of the game (assuming the opponent plays optimally as well).
- Can be computed using complete mini-max search of the finite game tree.

Approximating *V*(*b*)

- Computing V(b) is intractable since it involves searching the complete exponential game tree.
- Therefore, this definition is said to be **non- operational**.
- An **operational** definition can be computed in reasonable (polynomial) time.
- Need to learn an operational approximation to the ideal evaluation function.

Representing the Target Function

- Target function can be represented in many ways: lookup table, symbolic rules, numerical function, neural network.
- There is a trade-off between the expressiveness of a representation and the ease of learning.
- The more expressive a representation, the better it will be at approximating an arbitrary function; however, the more examples will be needed to learn an accurate function.

Linear Function for Representing *V*(*b*)

- In checkers, use a linear approximation of the evaluation function.

$$V(b) = w_0 + w_1 \cdot bp(b) + w_2 \cdot rp(b) + w_3 \cdot bk(b) + w_4 \cdot rk(b) + w_5 \cdot bt(b) + w_6 \cdot rt(b)$$

  - *bp*(*b*): number of black pieces on board b
  - *rp*(*b*): number of red pieces on board b
  - *bk*(*b*): number of black kings on board b
  - *rk*(*b*): number of red kings on board b
  - *bt*(*b*): number of black pieces threatened (i.e. which can be immediately taken by red on its next turn)
  - *rt*(*b*): number of red pieces threatened

Obtaining Training Values

- Direct supervision may be available for the target function.
  - < <*bp*=3,*rp*=0,*bk*=1,*rk*=0,*bt*=0,*rt*=0>, 100> (win for black)
- With indirect feedback, training values can be estimated using ***temporal difference learning*** (used in ***reinforcement learning*** where supervision is ***delayed reward***).

Temporal Difference Learning

- Estimate training values for intermediate (non- terminal) board positions by the estimated value of their successor in an actual game trace.

$$V_{train}(b) = V(\text{successor}(b))$$

where successor(b) is the next board position where it is the program's move in actual play.

- Values towards the end of the game are initially more accurate and continued training slowly "backs up" accurate values to earlier board positions.

Learning Algorithm

- Uses training values for the target function to induce a hypothesized definition that fits these examples and hopefully generalizes to unseen examples.
- In statistics, learning to approximate a continuous function is called ***regression***.
- Attempts to minimize some measure of error (***loss function***) such as ***Mean Squared Error (MSE)***:

$$MSE = \frac{1}{n}\sum_{i=1}^{n}(Y_i - \widehat{Y}_i)^2$$

- n is the number of observations in the dataset.
- Y_i is the actual value of the observation
- $\widehat{Y}_i$ is the predicted value of the ith observation.

Least Mean Squares (LMS) Algorithm

- A gradient descent algorithm that incrementally updates the weights of a linear function in an attempt to minimize the mean squared error
- Until weights converge :
  - For each training example *b* do :
    - » Compute the absolute error :

$$error(b) = V_{train}(b) - V(b)$$

    - » For each board feature, f_i, update its weight, w_i :

$$w_i = w_i + c \cdot f_i \cdot error(b)$$

for some small constant (learning rate) c

LMS Discussion

- Intuitively, LMS executes the following rules:
  - If the output for an example is correct, make no change.
  - If the output is too high, lower the weights proportional to the values of their corresponding features, so the overall output decreases
  - If the output is too low, increase the weights proportional to the values of their corresponding features, so the overall output increases.
- Under the proper weak assumptions, LMS can be proven to eventually converge to a set of weights that minimizes the mean squared error.

Lessons Learned about Learning

- Learning can be viewed as using direct or indirect experience to approximate a chosen target function.
- Function approximation can be viewed as a search through a space of hypotheses (representations of functions) for one that best fits a set of training data.
- Different learning methods assume different hypothesis spaces (representation languages) and/or employ different search techniques.

Various Function Representations

- Numerical functions
  - Linear regression
  - Neural networks
  - Support vector machines
- Symbolic functions
  - Decision trees
  - Rules in propositional logic
  - Rules in first-order predicate logic
- Instance-based functions
  - Nearest-neighbor
  - Case-based
- Probabilistic Graphical Models
  - Naïve Bayes
  - Bayesian networks
  - Hidden-Markov Models (HMMs)
  - Probabilistic Context Free Grammars (PCFGs)
  - Markov networks

Various Search Algorithms

- Gradient descent
  - Perceptron
  - Backpropagation
- Dynamic Programming
  - HMM Learning
  - PCFG Learning
- Divide and Conquer
  - Decision tree induction
  - Rule learning
- Evolutionary Computation
  - Genetic Algorithms (GAs)
  - Genetic Programming (GP)
  - Neuro-evolution

Evaluation of Learning Systems

- Experimental
  - Conduct controlled cross-validation experiments to compare various methods on a variety of benchmark datasets.
  - Gather data on their performance, e.g. test accuracy, training-time, testing-time.
  - Analyze differences for statistical significance.
- Theoretical
  - Analyze algorithms mathematically and prove theorems, based on:
    - Computational complexity
    - Ability to fit training data
    - Sample complexity (number of training examples needed to learn an accurate function)

History of Machine Learning

- 1950s
  - Samuel's checker player
  - Selfridge's Pandemonium
- 1960s:
  - Neural networks: Perceptron
  - Pattern recognition
  - Learning in the limit theory
  - Minsky and Papert prove limitations of Perceptron
- 1970s:
  - Symbolic concept induction
  - Winston's arch learner
  - Expert systems and the knowledge acquisition bottleneck
  - Quinlan's ID3
  - Michalski's AQ and soybean diagnosis
  - Scientific discovery with BACON
  - Mathematical discovery with AM

History of Machine Learning

- 1980s:
  - Advanced decision tree and rule learning
  - Explanation-based Learning (EBL)
  - Learning and planning and problem solving
  - Utility problem
  - Analogy
  - Cognitive architectures
  - Resurgence of neural networks (connectionism, backpropagation)
  - Valiant's PAC Learning Theory
  - Focus on experimental methodology
- 1990s
  - Data mining
  - Adaptive software agents and web applications
  - Text learning
  - Reinforcement learning (RL)
  - Inductive Logic Programming (ILP)
  - Ensembles: Bagging, Boosting, and Stacking
  - Bayes Net learning

History of Machine Learning

- 2000s
  - Support vector machines
  - Kernel methods
  - Graphical models
  - Statistical relational learning
  - Transfer learning
  - Sequence labeling
  - Collective classification and structured outputs
  - Computer Systems Applications
    - Compilers
    - Debugging
    - Graphics
    - Security (intrusion, virus, and worm detection)
  - E mail management
  - Personalized assistants that learn
  - Learning in robotics and vision

THANK YOU